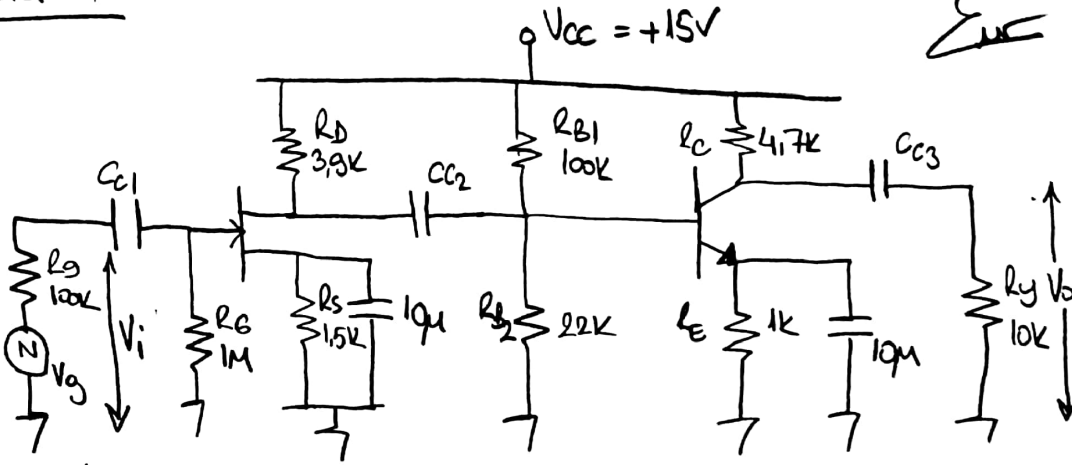


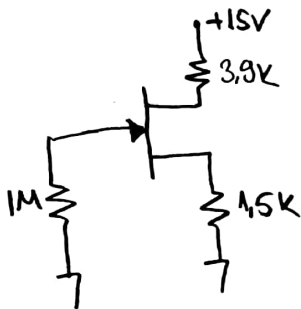
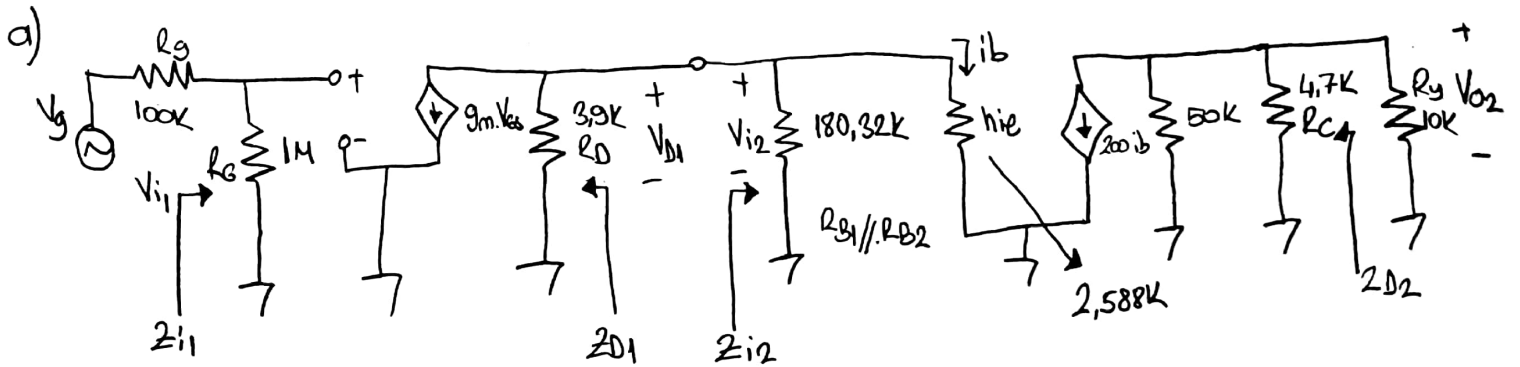
Cevaplar 1

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$I_{DSS} = 6mA$, $V_p = -3V$
 $h_{fe} = h_{FE} = 200$, $V_{BE} = 0.7V$
 $1/h_{oe} = 50k$, $C_{gs} = 3pF$
 $C_{ds} = 1pF$, $C_{gd} = 3pF$
 $C_{be} = 80pF$, $C_{bc} = 3pF$
 $C_{ce} = 6pF$

- a) $K_V = \frac{V_o}{V_i} = ?$ b) Alt kesim frekansının 10Hz olabilmesi için $C_{C1}, C_{C2}, C_{C3} = ?$
 c) Üst kesim frekansını nedir?



$$V_G = 0$$

$$V_G = V_{GS} + I_D \cdot 1.5k$$

$$I_D = \frac{V_{GS}}{1.5k}$$

$$I_D = I_{DSS} \cdot \left(1 - \frac{V_{GS}}{V_p}\right)^2 = 6m \cdot \left(1 - \frac{V_{GS}}{-3}\right)^2$$

$$I_D = 6m \left(1 + \frac{2}{3}V_{GS} + \frac{V_{GS}^2}{9}\right)$$

$$I_D = 6m \left(\frac{9 - 6V_{GS} + V_{GS}^2}{9}\right) = \frac{6m}{9} (9 - 6V_{GS} + V_{GS}^2)$$

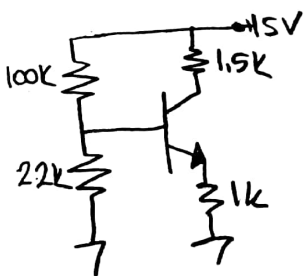
$$-\frac{V_{GS}}{1.5k} = \frac{6m}{9} (9 - 6V_{GS} + V_{GS}^2)$$

$$-9V_{GS} = 1.5k \cdot 6m (9 - 6V_{GS} + V_{GS}^2)$$

$$0 = 81 + 63V_{GS} + 9V_{GS}^2$$

$$V_{GS} = -1.7V$$

$$A_{V1} = g_m \cdot (R_D \parallel Z_{i2}) = 1.73m \cdot (3.9k \parallel 2.26k) \quad A_{V1} = 2.47$$



$$r_e = \frac{26mV}{I_{E1}}$$

$$r_e = \frac{26mV}{2.01mA}$$

$$r_e = 12.95\Omega$$

$$V_B = 15 \cdot \frac{22k}{22k + 100k}$$

$$V_B = 2.72V$$

$$Z_{i2} = 18.03 \parallel 2.58k = 2.26k$$

$$3.9k \parallel 2.26k = 1.44k$$

$$V_B = V_{BE} + I_E \cdot 1k$$

$$2.72 = 0.7 + I_E \cdot 1k$$

$$I_{E1} = \frac{2.72 - 0.7}{1k} = 2.02mA$$

$$h_{ie} = B \cdot r_e = h_{fe} \cdot r_e = 200 \cdot 12,95 \Omega = \underline{2,590k\Omega}$$

$$A_{V2} = \frac{V_{o2}}{V_{i2}} = -200 \cdot b \cdot \left(\frac{50k // 4,7k // 10k}{b \cdot 2,590k} \right) \rightarrow \underline{3k}$$

$$= \frac{-200 \cdot 3k}{b \cdot 2,590k} = -231,66 = A_{V2}$$

$$K_V = A_{V1} A_{V2} = -2,47 \cdot -231,66$$

$$\underline{K_V = 572,2}$$

b) C_{c1} ;

$$f_{c1} = \frac{1}{2\pi \cdot C_{c1} \cdot R_{eq}} \Rightarrow 10 = \frac{1}{2\pi \cdot C_{c1} \cdot 1,1M} = C_{c1} = \frac{1}{2\pi \cdot 10 \cdot 1,1M} \Rightarrow \underline{C_{c1} = 144,68 \mu F}$$

$R_{eq} = R_g // R_6 = 1,1M$

C_{c2}

$$f_{c2} = \frac{1}{2\pi \cdot C_{c2} \cdot R_{eq}} \Rightarrow 10 = \frac{1}{2\pi \cdot C_{c2} \cdot 6,16k} \Rightarrow C_{c2} = \frac{1}{2\pi \cdot 10 \cdot 6,16k} \Rightarrow \underline{C_{c2} = 2,58 \mu F}$$

$R_{eq} = 3,9k + Z_{i2} = 6,16k$

C_{c3}

$$f_{c3} = \frac{1}{2\pi \cdot C_{c3} \cdot R_{eq}} = 10 = \frac{1}{2\pi \cdot C_{c3} \cdot 14,3k} \Rightarrow C_{c3} = \frac{1}{2\pi \cdot 10 \cdot 14,3k} \Rightarrow \underline{C_{c3} = 1,11 \mu F}$$

$R_{eq} = (50k // 4,7k) + 10k = 14,3k$

c) FET;

$$C_i; f_{Hi} = \frac{1}{2\pi \cdot C_i \cdot R_{THi}}$$

$$R_{THi} = R_g // R_G = 100k // 1M = 90,90k\Omega$$

$$C_i = \overset{\circ}{C_{wi}} + C_{gs} + (1 - A_{V1}) \cdot C_{gd} = 3pF + (1 - (-2,47)) \cdot 3pF$$

$$= \frac{1}{2\pi \cdot 13,41pF \cdot 90,90k\Omega}$$

$$\underline{C_i = 13,41pF}$$

$$\underline{f_{Hi} = 130,55 kHz}$$

$$C_o; f_{Ho} = \frac{1}{2\pi \cdot C_o \cdot R_{THo}}$$

$$R_{THo} = R_o // Z_{i2} = 3,9k // 2,26k = 1,73k\Omega$$

$$C_o = \overset{\circ}{C_{wo}} + C_{ds} + \left(1 - \frac{1}{A_{V1}}\right) \cdot C_{gd} = 1pF + \left(1 - \frac{1}{(-2,47)}\right) \cdot 3pF$$

$$= \frac{1}{2\pi \cdot 5,22pF \cdot 1,73k\Omega}$$

$$\underline{C_o = 5,22pF}$$

$$\underline{f_{Ho} = 17,63 MHz}$$

$$C_i; f_{Hi} = \frac{1}{2\pi \cdot C_i \cdot R_{THi}}$$

$$R_{THi}, R_o // Z_{i2} = 1,73k\Omega$$

$$C_i = C_{wi} + C_{be} + (1 - A_{V2}) \cdot C_{bc} = 80pF + (1 - (-232,23)) \cdot 3pF$$

$$= \frac{1}{2\pi \cdot 776,69pF \cdot 1,73k\Omega}$$

$$\underline{C_i = 776,69pF}$$

$$\underline{f_{Hi} = 118,45 kHz}$$

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$$C_0 ; f_{H0} = \frac{1}{2\pi \cdot C_0 \cdot R_{TH0}}$$
$$= \frac{1}{2\pi \cdot 9,01 \text{ pF} \cdot 3 \text{ k}\Omega}$$

$$\underline{\underline{f_{H0} = 5,88 \text{ MHz}}}$$

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$$R_{TH0} = 50 \text{ k}\Omega // 4,7 \text{ k}\Omega // 10 \text{ k}\Omega = 3 \text{ k}\Omega$$

$$C_0 = C_{wo} + C_{ce} + \left(1 - \frac{1}{A_v}\right) \cdot C_{be} = 6 \text{ pF} + \left(1 + \frac{1}{22,33}\right) 3 \text{ pF}$$

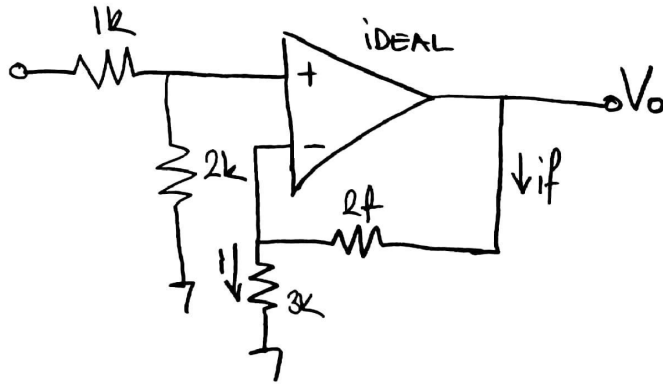
$$\underline{\underline{C_0 = 9,01 \text{ pF}}}$$

$$\underline{\underline{f_{gst} = 118,45 \text{ kHz}}}$$

Gözümler 2

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Aşağıda verilen ideal OPAMP'ın kazançının 5 olabilmesi için R_f ne olmalıdır?



$$\text{ideal OPAMP} = V_+ = V_-$$

$$AV = \frac{V_o}{V_i} = 5$$

$$V_+ = V_i = \frac{2k}{2k+1k} \cdot V_o$$

$$V_- = V_o \cdot \frac{3k}{3k+R_f}$$

$$V_i \cdot \frac{2k}{3k} = V_o \cdot \frac{3k}{3k+R_f}$$

$$2k(3k+R_f) = \underbrace{\frac{V_o}{V_i}}_5 \cdot 3k \cdot 3k$$

$$6M + 2k \cdot R_f = 5 \cdot 9M$$

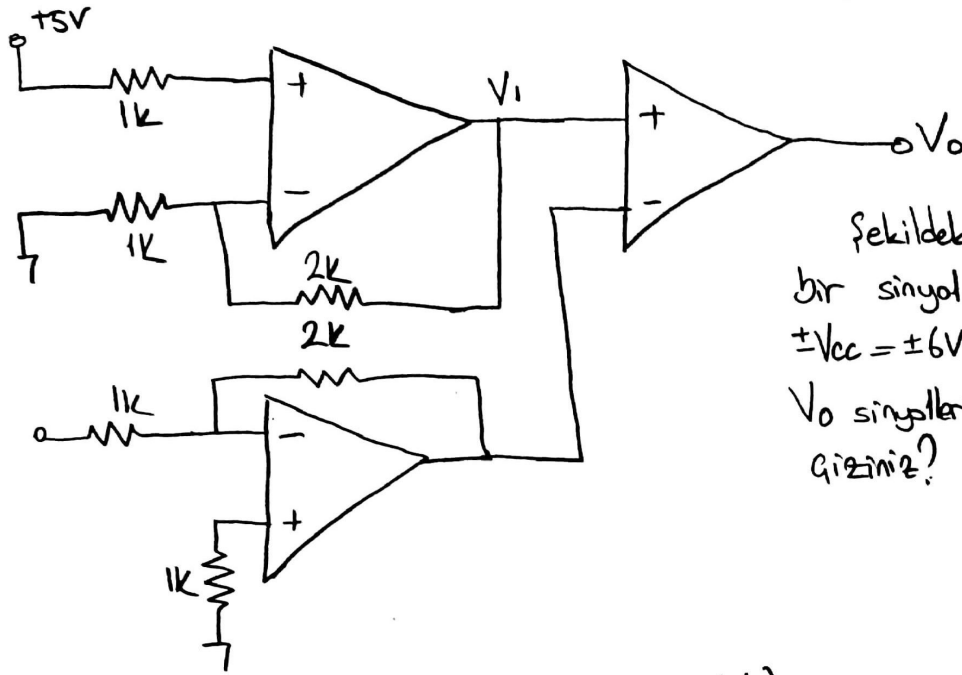
$$2k \cdot R_f = 45M - 6M$$

$$R_f = \frac{39M}{2k}$$

$$\underline{\underline{R_f = 19,5k\Omega}}$$

Görüntüler 3

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Şekildeki devrede $V_1 \pm 1V$ sinüoidal bir sinyal ve tüm Opomplar için $\pm V_{cc} = \pm 6V$ olduğuna göre V_1, V_2 ve V_o sinyallerini V_1 'ye göre ölçekli olarak çiziniz?

$$V_- = V_+ = 5V$$

$$V_- = V_1 \cdot \frac{1k}{1k + 2k}$$

$$V_1 = \frac{5 \cdot 3k}{1k} = 15V$$

V_{cc} 'den büyük olduğu için $+V_{cc}$ dir

$$\underline{\underline{V_1 = 6V}}$$

$$V_+ = V_- = 0V$$

$$\frac{V_1 - V_1}{1k} = \frac{V_- - V_2}{2k}$$

$$\frac{V_1}{1k} = -\frac{V_2}{2k}$$

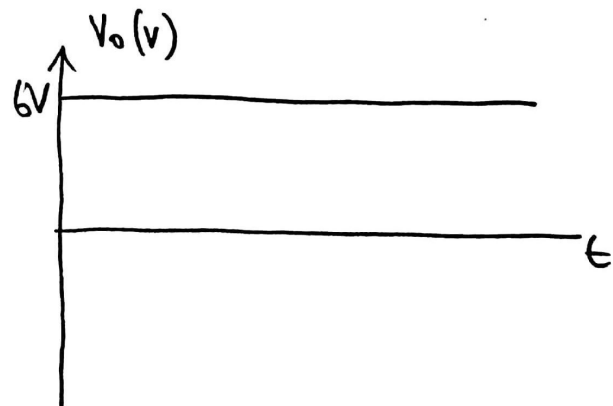
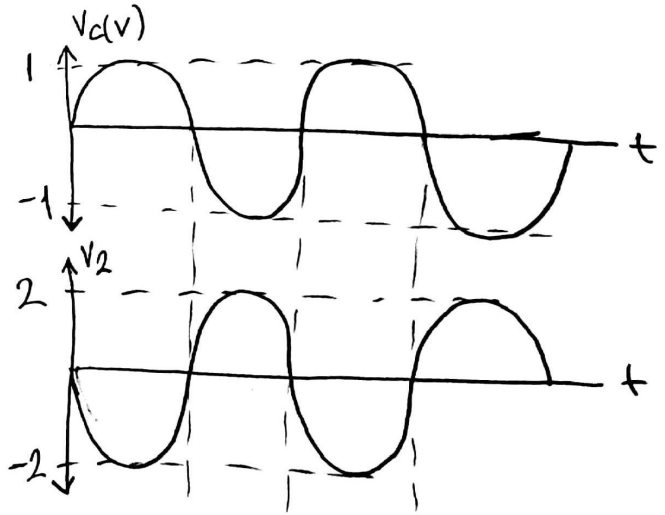
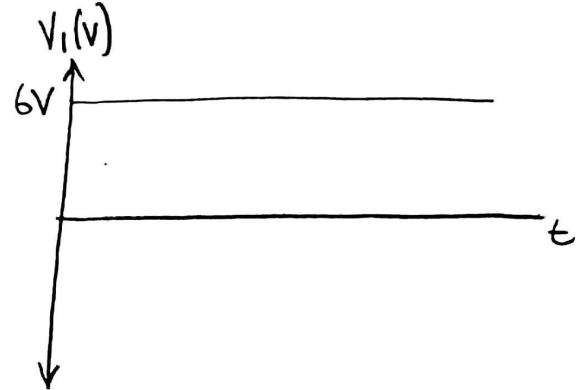
$$2V_1 = -V_2$$

$$V_2 = -2V_1$$

$$V_1 = V_+ = 6V$$

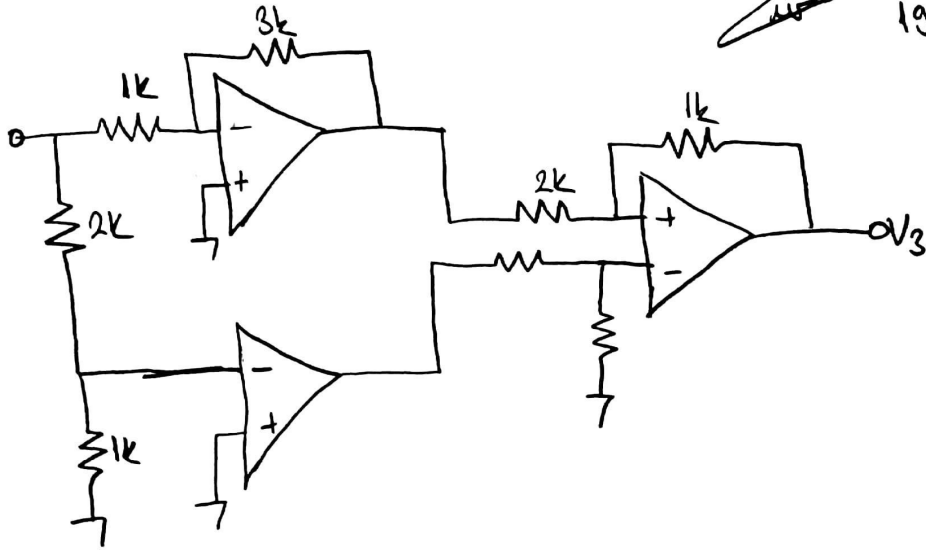
$$V_2 = V_- = -2V_1$$

V_+ girişi V_- girişinden büyük olduğu için $+V_{cc}$ dir



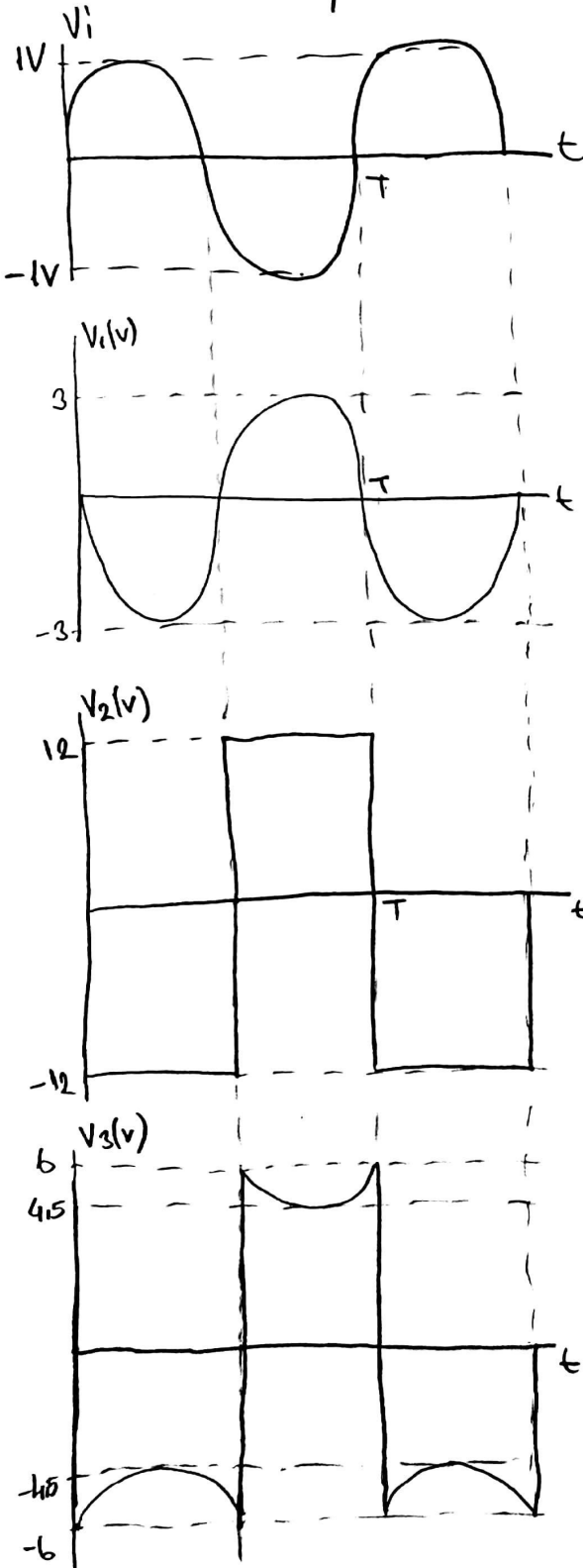
Cevaplar 4

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$$\pm V_{cc} = \pm 12V$$

Buna göre $V_1, V_2, V_3 = ?$ Çiziniz.



$$V_+ = V_- = 0$$

$$\frac{V_1 - \hat{V}_-}{1k} = \frac{\hat{V}_- - V_1}{3k}$$

$$V_1 = -3V_i$$

$$V_- = V_i \cdot \frac{1k}{1k+2k} = \frac{V_i}{3}$$

$$V_- = 0.33V_i$$

$$\left. \begin{array}{l} V_i > 0 \quad -V_{cc} = -12V \\ V_i < 0 \quad +V_{cc} = +12V \end{array} \right\} V_2$$

$$V_1 \text{ iken } \Rightarrow (V_2 = 0, V_- = 0)$$

$$\frac{V_1 - 0}{2k} = \frac{0 - V_{31}}{1k} \quad \underline{\underline{V_{31} = -\frac{V_1}{2}}}$$

$$V_2 \text{ iken } \Rightarrow (V_1 = 0)$$

$$V_+ = V_{32} \cdot \frac{2k}{2k+1k} \quad V_- = V_2 \cdot \frac{1k}{1k+2k}$$

$$V_+ = V_-$$

$$V_{32} \cdot \frac{2}{3} = V_2 \cdot \frac{1}{3}$$

$$\underline{\underline{V_{32} = \frac{V_2}{2}}}$$

$$V_3 = V_{31} + V_{32}$$

$$\underline{\underline{V_3 = \frac{V_2 - V_1}{2}}}$$

$$V_3(-1) = \frac{-12 - (0)}{2} = -6V //$$

$$V_3(-2) = \frac{-12 - (-3)}{2} = -4.5V //$$

$$V_3(+1) = \frac{12 - 0}{2} = 6V //$$

$$V_3(+2) = \frac{12 - 3}{2} = 4.5V //$$